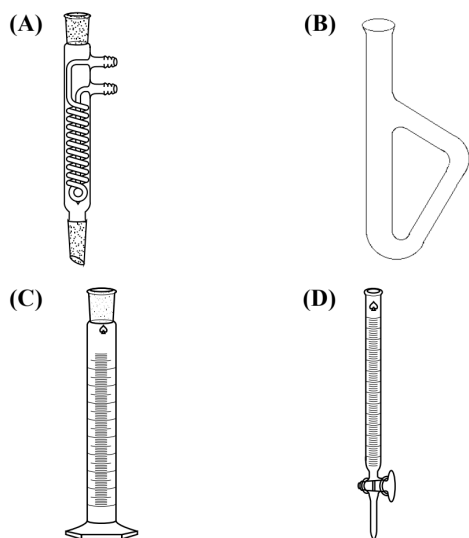


Descriptive/Laboratory

1. A gas mixture extinguishes a glowing splint and does not cause limewater to turn cloudy when bubbled through it. Which most accurately describes the gas mixture?

(A) It contains only carbon dioxide.
(B) It contains only oxygen.
(C) It contains both carbon dioxide and oxygen.
(D) It contains neither carbon dioxide nor oxygen.

2. Which laboratory glassware is most useful for determining the melting point of a solid organic compound?



3. Which gas can be condensed to a blue liquid?

(A) Carbon dioxide (B) Hydrogen
(C) Nitrogen dioxide (D) Ozone

4. Which liquids are miscible in all proportions?

(A) H_2O and CH_2Cl_2 (B) H_2O and $(\text{CH}_3)_2\text{CO}$
(C) C_6H_{12} and Hg (D) C_6H_{12} and CH_3OH

5. Which is the best way to prepare 0.5 L of a 1 M solution of sulfuric acid?

(A) 0.5 mol of concentrated sulfuric acid should be added to a 500 mL volumetric flask, followed by enough distilled water to fill the flask to the mark.
(B) 0.5 mol of concentrated sulfuric acid should be added slowly to about 300 mL of distilled water in a 1 L Erlenmeyer flask, then enough distilled water should be added to reach the 500 mL mark on the flask.
(C) 11.2 L of sulfur trioxide gas, measured at STP, should be bubbled through 500 mL of distilled water in a 1 L Erlenmeyer flask.
(D) To 0.5 mol of silver sulfate suspended in about 300 mL water in a 500 mL volumetric flask should be added 1.0 mol HCl (as a concentrated aqueous solution). The solution should be diluted to the mark with distilled water and then filtered to remove precipitated AgCl.

6. A chemist determines the fraction of magnesium that burns in air to make Mg_3N_2 by adding the ash formed from a known mass of Mg ribbon into water, distilling the mixture into a known excess of hydrochloric acid, and back-titrating the excess HCl with NaOH to a methyl orange endpoint. Which error will cause the calculated fraction of magnesium nitride to be too high?

(A) The distillate is collected in a flask by itself and subsequently added to the HCl.
(B) Phenolphthalein is used as an indicator in place of methyl orange.
(C) The potassium hydrogen phthalate used to standardize the NaOH had absorbed some moisture from the air.
(D) When the HCl solution was drawn into the volumetric pipet used to dispense it, a large bubble was present.

7. A white solid is either sodium carbonate or sodium sulfate. Which procedure will best distinguish between the two?

(A) Adding 1.0 M hydrochloric acid to the two solids
(B) Adding 1.0 M barium nitrate to solutions made from the two solids
(C) Adding 1.0 M potassium bromide to solutions made from the two solids
(D) Measuring the conductivity of solutions made from the two solids

8. A colorimeter contains four LEDs. Which one would give the best results in determining the concentration of a substance that forms a blue solution?

(A) Violet (430 nm) (B) Blue (470 nm)
(C) Green (565 nm) (D) Red (635 nm)

Descriptive/Laboratory

9. A student wishes to determine the number of colored dyes present in the ink of a black felt-tip marker. Which technique would be most suitable?

(A) Paper chromatography
(B) Gas chromatography
(C) Nuclear magnetic resonance spectroscopy
(D) Ultraviolet-visible spectroscopy

10. A solution of a metal chloride gives a red flame test. Which metal is it?

(A) Na (B) K (C) Cu (D) Sr

11. Exposure of an aqueous solution of which ion to oxygen gas will cause a color change?

(A) Cr^{2+} (B) $\text{Cr}_2\text{O}_7^{2-}$
(C) $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ (D) $\text{Fe}(\text{CN})_6^{3-}$

12. A student determines the rate constant for the reaction of iodide with hydrogen peroxide in strongly acidic solution ($\text{Rate} = k[\text{H}_2\text{O}_2][\text{I}^-][\text{H}^+]$) by mixing 1.00 mL aliquots of solutions of known concentrations of potassium iodide, ascorbic acid, and hydrochloric acid with four drops of starch solution, then adding 1.00 mL of hydrogen peroxide solution and recording the time required for the solution to turn blue. A student uses half the stipulated amount of one of the solutions but does not realize it. Underdosing of which solutions will cause the value of the rate constant to be too high?

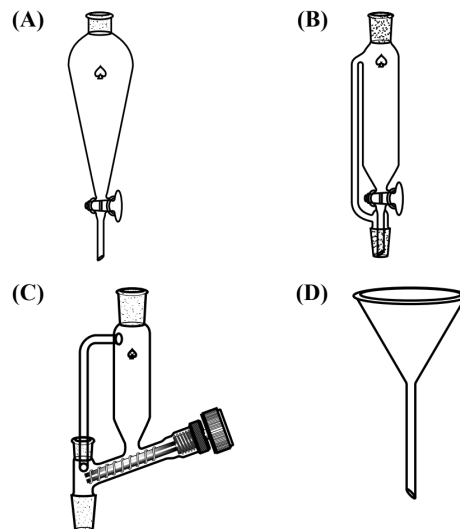
I. Hydrochloric acid II. Ascorbic acid

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

13. An unknown alcohol is thought to be either 2-pentanol or 3-pentanol. Which instrumental method would most effectively distinguish between these two possibilities?

(A) Mass spectrometry
(B) Infrared spectroscopy
(C) Ultraviolet spectroscopy
(D) ^{13}C NMR spectroscopy

14. A neutral organic compound is to be separated from a basic impurity by washing a solution of the compound in hexane with 5% aqueous HCl. Which apparatus is best suited to this operation?



15. Which substance may be treated with bleach (sodium hypochlorite) to render it less hazardous?

(A) Sodium cyanide (B) Ammonia
(C) Hydrochloric acid (D) Ozone

16. Which indicator would be most suitable for a titration to determine the concentration of ammonia in a window cleaning solution using aqueous HCl?

(A) Eriochrome Black T (B) Ferroin
(C) Methyl orange (D) Phenolphthalein

17. A test tube contains 4 mL of a solution that is 0.1 M in one or more of the following salts: AgNO_3 , $\text{Mn}(\text{NO}_3)_2$, and $\text{Al}(\text{NO}_3)_3$. 1 mL of 6 M HCl is added to the test tube, which causes neither a color change nor formation of a precipitate. After subsequent addition of 1.5 mL aqueous NaOH and 0.5 mL of 3% aqueous H_2O_2 and brief heating in a hot water bath, there is a dark precipitate in the test tube. What may be determined about the cations initially present?

	Must be present	May be present	Must be absent
(A)	Ag^+	Al^{3+}	Mn^{2+}
(B)	Al^{3+}	Mn^{2+}	Ag^+
(C)	Al^{3+}	Ag^+	Mn^{2+}
(D)	Mn^{2+}	Al^{3+}	Ag^+

Descriptive/Laboratory

18. Ethane, C_2H_6 , and diborane, B_2H_6 , do not adopt analogous molecular structures. Which experimental technique would be least useful in establishing this fact?
- (A) Infrared spectroscopy
(B) Mass spectrometry
(C) Nuclear magnetic resonance spectroscopy
(D) X-ray crystallography
19. The concentration of a solution of potassium permanganate is determined by titrating it into an Erlenmeyer flask with solid ferrous ammonium sulfate hexahydrate ($Fe(NH_4)_2(SO_4)_2 \cdot 6 H_2O$) dissolved in dilute sulfuric acid. Which error would result in a measured concentration of $KMnO_4$ that is lower than the actual concentration?
- (A) The titration was begun before the ferrous ammonium sulfate had fully dissolved in the sulfuric acid.
(B) The sides of the Erlenmeyer flask were periodically rinsed down with deionized water during the titration.
(C) Some of the $Fe(II)$ ions in the solid ferrous ammonium sulfate had been oxidized to $Fe(III)$ on storage.
(D) The buret was rinsed with distilled water and then filled with $KMnO_4$ solution while still wet.
20. A weighed sample of iron ore is dissolved in 10 mL of 1.0 M hydrochloric acid and then titrated with 0.100 M $KMnO_4$ to determine the percent iron by mass in the sample. Which glassware is the best choice for measuring the hydrochloric acid?
- (A) 50 mL buret
(B) 10 mL volumetric pipet
(C) 10 mL graduated pipet
(D) 10 mL graduated cylinder
21. A 0.1 M aqueous solution containing which metal ion is least intensely colored?
- (A) Cr^{2+} (B) Mn^{2+} (C) Co^{2+} (D) Ni^{2+}
22. When equal volumes of a 1.0 M ammonium sulfate solution and a saturated solution of barium hydroxide are mixed, what is observed?
- I. Gas is evolved
II. A white precipitate forms.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
23. A weighed sample of a copper-nickel alloy is dissolved in a known volume of nitric acid. Which method is most suitable for determining the mass percent of copper in the alloy?
- (A) Treatment of an aliquot of the solution with excess iodide, followed by titration of the iodine produced with sodium thiosulfate.
(B) Measurement of the absorbance of the solution at a wavelength of light at which both Cu^{2+} and Ni^{2+} absorb, and comparison with the absorbances of known standards of the two ions.
(C) Addition of excess sodium hydroxide to the solution, isolation of the metal hydroxides by filtration, and measurement of the mass of the precipitate.
(D) Bubbling hydrogen gas through the solution and measuring the mass of the metal that precipitates from the solution.
24. Which metal ion forms the least soluble perchlorate salt?
- (A) Li^+ (B) Mg^{2+} (C) K^+ (D) Ca^{2+}
25. A sample of malachite, $Cu_2CO_3(OH)_2$, is heated over a Bunsen burner to form CuO . The mass percent copper in the sample is calculated based on the mass before and after heating. Which could explain the experimentally determined percent copper being lower than that expected for malachite?
- (A) The sample is not heated as strongly as recommended in the procedure (not long or hot enough).
(B) The sample is heated more strongly than recommended in the procedure (too long or too hot).
(C) Azurite ($Cu_3(CO_3)_2(OH)_2$) is present in the sample.
(D) Goethite ($FeO(OH)$) is present in the sample.
26. Which element gives a red flame test?
- (A) Aluminum (B) Boron
(C) Sodium (D) Strontium
27. Combining which solutions would produce a color change?
- (A) $NaCl(aq)$ and $Br_2(aq)$
(B) $Na_2CO_3(aq)$ and $HNO_3(aq)$
(C) $K_2C_2O_4(aq)$ and $FeCl_3(aq)$
(D) $CH_3COOH(aq)$ and $Co(NO_3)_2(aq)$
28. Which metal resists oxidation in air because it forms a protective oxide coating?
- (A) Al (B) Ca (C) Fe (D) Sr

Descriptive/Laboratory

29. Which experiment would be most appropriate for determining the number of components in a commercial nail polish remover?
- (A) Gas chromatography
(B) Paper chromatography
(C) Boiling point determination
(D) Combustion analysis
30. A solution containing a divalent metal ion forms a precipitate when treated with hydrogen sulfide which does not dissolve in 1 M HCl. Which ion is present?
- (A) Ca^{2+} (B) Mn^{2+} (C) Cd^{2+} (D) Ba^{2+}
31. An unknown metal M reacts with aqueous sulfuric acid to form hydrogen gas and the divalent metal ion $\text{M}^{2+}(\text{aq})$. A student collects the hydrogen over water and calculates the molar mass of M based on the volume of hydrogen collected. Which error will result in a value of the calculated molar mass that is higher than the true molar mass?
- (A) The concentration of sulfuric acid is 2.0 M rather than 1.0 M as recommended by the procedure.
(B) Some of the metal sample has oxidized to the metal oxide.
(C) The barometric pressure is assumed to be 760 torr when in fact it is 740 torr.
(D) The vapor pressure of water is not included in the calculation.
32. Which glassware would be most appropriate for measuring 10.00 mL of distilled water?
- (A) A 10-mL test tube
(B) A 25-mL beaker
(C) A 50-mL buret
(D) A 100-mL graduated cylinder
33. A solution that may contain either 0.1 M $\text{Ag}^+(\text{aq})$, 0.1 M $\text{Pb}^{2+}(\text{aq})$, or both, is treated with 1 M aqueous HCl. A white precipitate forms which does not appear to dissolve in hot water. Which conclusion about the cations present may be drawn?
- (A) Only Ag^+ is present.
(B) Only Pb^{2+} is present.
(C) Ag^+ is present, and Pb^{2+} may be present.
(D) Pb^{2+} is present, and Ag^+ may be present.
34. Addition of 10 mL of distilled water to 0.1 g of which salt produces a yellow, slightly cloudy solution?
- (A) KMnO_4 (B) FeCl_3
(C) $\text{Co}(\text{NO}_3)_2$ (D) CuSO_4
35. The amount of ascorbic acid in a vitamin C tablet is determined by titration with a solution of iodine in aqueous potassium iodide. A small amount of starch is added to the vitamin C solution before the titration. What is the function of the starch?
- (A) It forms an intensely colored complex with triiodide ion.
(B) It increases the viscosity of the analyte solution.
(C) It catalyzes the dissociation of triiodide into iodine and iodide.
(D) It binds to the inert ingredients of the vitamin C tablet.
36. When blue solid $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$ is heated, it turns white. Which is the best explanation for this observation?
- (A) Heating causes water molecules bonded to copper to be replaced by sulfate, lowering the energy of the light absorbed.
(B) Heating causes an expansion of the lattice, leading to a decrease in the coordination number of copper and raising the energy of the light absorbed.
(C) Heating causes reduction of copper(II) to copper(I).
(D) Heating causes reduction of sulfate to sulfite.
37. A student performs an experiment to determine the concentration of a colored salt solution by measuring the absorbance of the solution at the wavelength of maximum absorbance of the salt (λ_{max}) and using Beer's Law to calculate the concentration. Which of the following could cause the measured concentration to be higher than the actual concentration?
- (A) The cuvette is not rinsed with the salt solution after being washed.
(B) The cuvette is not wiped off before it is inserted into the spectrophotometer.
(C) Less than the recommended volume of salt solution is added to the cuvette.
(D) The spectrometer is set to a wavelength different from λ_{max} .
38. Which ion gives a green flame test?
- (A) Potassium (B) Calcium
(C) Strontium (D) Barium

Descriptive/Laboratory

39. The equilibrium constant for the formation of CoCl_4^{2-} , a species that is blue in solution, is to be measured using a colorimeter. The colorimeter has wavelength settings of 470 nm ("blue"), 565 nm ("green"), and 635 nm ("red") to use in the experiment. What is the best setting to use?

(A) 470 nm
(B) 565 nm
(C) 635 nm
(D) All settings would be equally suitable for the measurement.

40. A student is using a coffee-cup calorimeter to determine the enthalpy change of the endothermic reaction of two aqueous solutions. After both solutions are added to the cup, the student neglects to put the lid on the cup. This would cause the magnitude of the calculated ΔH° value to be:

(A) too small, since some heat will escape out of the cup.
(B) too large, since some heat will escape out of the cup.
(C) too small, since the solution will absorb heat from the room.
(D) too large, since the solution will absorb heat from the room.

41. Which substance produces a toxic and explosive gas when added to strong acids?

(A) NaN_3 (B) Na_2CO_3
(C) NaClO_4 (D) Na_2SO_3

42. When aqueous ethanol is treated with an acidified solution of potassium dichromate, what is observed?

I. Color change
II. Gas evolution

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

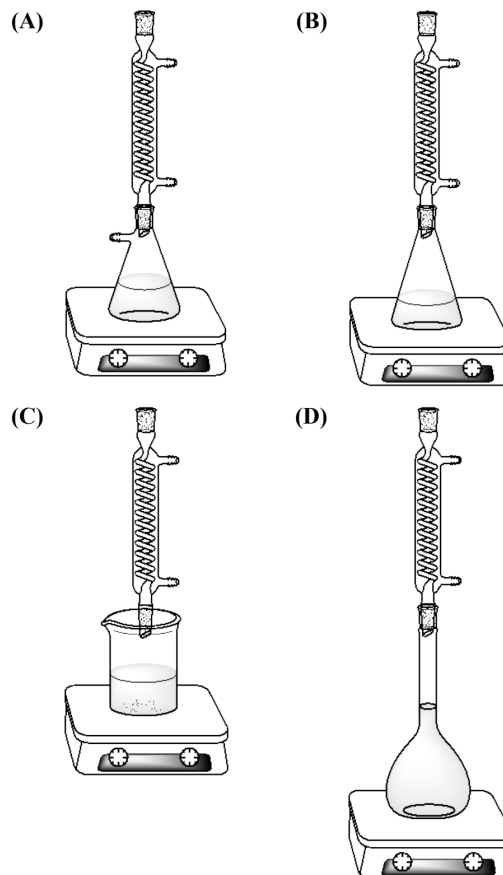
43. Mixing which 0.1 M aqueous solutions results in formation of a colored precipitate?

(A) BaCl_2 and CH_3COOH
(B) BaCl_2 and Na_2CO_3
(C) CuCl_2 and CH_3COOH
(D) CuCl_2 and Na_2CO_3

44. Which dissolves to give a blue solution?

(A) $\text{Na}_2\text{S}_2\text{O}_4$ (B) K_2CrO_4
(C) $\text{Cu}(\text{NO}_3)_2$ (D) $\text{Zn}(\text{OH})_2$

45. A chemist wishes to prepare methyl benzoate by refluxing benzoic acid in methanol overnight in the presence of a catalytic amount of sulfuric acid. Which setup is most appropriate for carrying out this reaction?



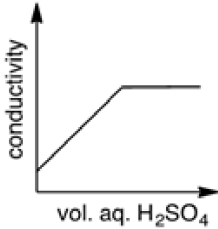
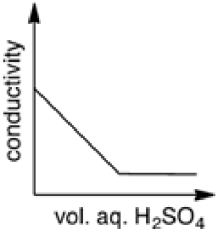
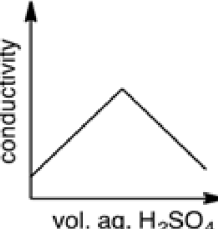
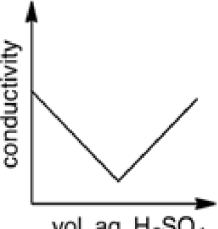
46. A pale yellow solid is insoluble in water or concentrated ammonia solution, but dissolves in concentrated $\text{Na}_2\text{S}_2\text{O}_3$ solution. What is it?

(A) AgF (B) AgCl
(C) AgI (D) $\text{Ag}(\text{CH}_3\text{COO})$

47. A chemist has isolated a solid carboxylic acid of unknown formula from a medicinal herb. Which experiment will be most useful in assessing the purity of the isolated material?

(A) Melting point
(B) Titration with NaOH
(C) Combustion analysis
(D) UV-visible spectrophotometry

Descriptive/Laboratory

48. The ammonia concentration of a solution is determined by titrating with aqueous HCl (previously standardized against Na_2CO_3) using a pH meter. Which of the following errors will lead to a measured concentration of NH_3 that is higher than the actual concentration?
- (A) Some of the Na_2CO_3 used in the standardization is spilled before being transferred to the titration flask.
- (B) The glass stirring rod used to stir the ammonia solution is wiped with a paper towel after each aliquot of HCl is added.
- (C) The ammonia solution is allowed to stand in an open beaker for an hour before being titrated.
- (D) The pH meter has been miscalibrated so that all readings are 2.00 pH units higher than the actual pH.
49. Which titration would NOT require the addition of an indicator?
- (A) Determination of oxalate ion by titration with permanganate ion in acidic solution
- (B) Determination of chloride ion by titration with silver ion
- (C) Determination of acetic acid by titration with sodium hydroxide
- (D) Determination of ammonia by titration with hydrochloric acid
50. A 1 mL sample of a 0.1 M solution of a divalent metal ion is pale pink. To it is added 1 mL of concentrated HCl, which causes a color change to bright blue. What is the metal ion?
- (A) Ca^{2+} (B) Mn^{2+} (C) Co^{2+} (D) Cu^{2+}
51. A copper-nickel alloy is analyzed by dissolving it in 8 M nitric acid, diluting the solution with water, and then adding 1 mL of this diluted solution to an excess of aqueous potassium iodide. What are the principal forms of copper and nickel in this final mixture?
- (A) $\text{Cu}^{2+}(\text{aq})$, $\text{Ni}^{3+}(\text{aq})$ (B) $\text{CuI}_2(\text{s})$, $\text{NiI}_2(\text{s})$
- (C) $\text{CuI}(\text{s})$, $\text{Ni}^{2+}(\text{aq})$ (D) $\text{Cu}^+(\text{aq})$, $\text{Ni}(\text{s})$
52. A dilute solution of sulfuric acid is titrated into a solution of barium hydroxide and the conductivity of the solution is measured as a function of the added volume of sulfuric acid. Which graph represents the results of this experiment?
- (A) 
- (B) 
- (C) 
- (D) 
53. Which is the best way to prepare 500 mL of a 2.00 M solution of aqueous H_2SO_4 from deionized water ($M = 18.02$, density = 1.00 g mL^{-1}) and concentrated H_2SO_4 ($M = 98.08$, density = 1.84 g mL^{-1})?
- (A) Weigh 98.1 g concentrated sulfuric acid into a 500-mL beaker, then slowly add deionized water to the beaker, with occasional swirling, until the liquid reaches the 500 mL mark.
- (B) Weigh 98.1 g concentrated sulfuric acid into a 500-mL volumetric flask, slowly add deionized water to the mark, and mix.
- (C) Weigh 98.1 g concentrated sulfuric acid into a 100-mL beaker, then slowly pour the H_2SO_4 into a 500-mL beaker with about 250 mL deionized water in it. Pour this solution into a 500-mL volumetric flask and fill to the mark with deionized water and mix.
- (D) Weigh 446.6 g deionized water into a 500-mL volumetric flask, fill to the mark with concentrated sulfuric acid, and mix.
54. When a solution of barium hydroxide is mixed with a solution of iron(III) chloride, what is observed?
- (A) Precipitation of a colored solid
- (B) Precipitation of a colorless solid
- (C) Evolution of a colorless gas
- (D) Neither precipitation nor gas evolution

Descriptive/Laboratory

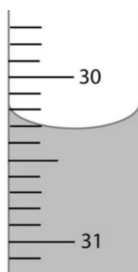
55. Which element is a liquid at 25 °C and 1 atm pressure?

- (A) Fluorine (B) Chlorine
(C) Bromine (D) Iodine

56. The concentration of which approximately 0.01 M solution could be most accurately determined by a visible spectrophotometer (or colorimeter)?

- (A) $\text{Mn}(\text{NO}_3)_2$ (B) $\text{Co}(\text{NO}_3)_2$
(C) $\text{Zn}(\text{NO}_3)_2$ (D) $\text{Pb}(\text{NO}_3)_2$

57. What is the reading of the buret shown?



- (A) 30.20 mL (B) 30.25 mL
(C) 30.30 mL (D) 31.75 mL

58. A student wishes to determine the molar mass of a pure solid organic compound. Which measurement would be most useful?

- (A) Melting point of the solid
(B) Melting point depression of a mixture of the solid with 1,4-dichlorobenzene
(C) Heat of combustion
(D) Solubility in water

59. A student has 10 mL of a solution that might contain any or all of the following cations at 0.01 M concentrations: Mn^{2+} , Ba^{2+} , Ag^{+} , and Cu^{2+} . Addition of 10 mL of 1 M HCl causes a precipitate to form. After the precipitate is filtered off, 1 M H_2SO_4 is added to the filtrate and another precipitate forms. What is the second precipitate?

- (A) MnSO_4
(B) BaSO_4
(C) Ag_2SO_4
(D) A mixture of BaSO_4 and Ag_2SO_4

60. When 6 M sodium hydroxide is added to an unknown white solid, the solid dissolves. What is a possible identity for this solid?

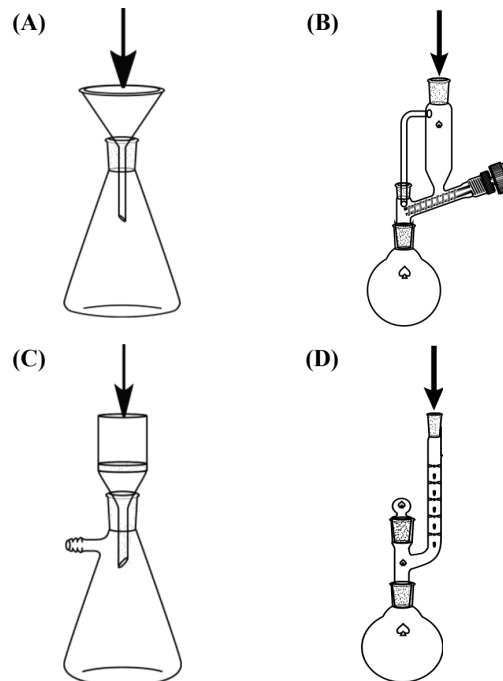
- (A) $\text{Mg}(\text{OH})_2$ (B) $\text{Al}_2(\text{SO}_4)_3$
(C) BaCO_3 (D) AgBr

61. A student standardizes a solution of aqueous NaOH against a measured mass of solid potassium hydrogen phthalate. She then uses this NaOH solution to titrate a measured mass of an unknown monocarboxylic acid to its phenolphthalein endpoint to determine its molar mass. Which errors will lead to a value of the molar mass that is too high?

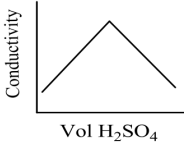
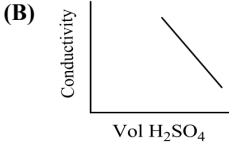
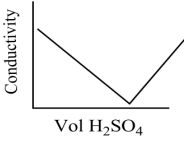
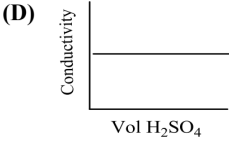
- I. The potassium hydrogen phthalate is partially hydrated.
II. The NaOH solution is allowed to stand after being standardized and absorbs some carbon dioxide from the air.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

62. A student has crystallized acetylsalicylic acid from a mixture of ethanol and water. Which apparatus is best suited for isolation of the crystalline material? (The point of addition of the solid/liquid mixture is indicated by the arrow.)



Descriptive/Laboratory

63. Which is the safest method for performing a flame test?
- (A) Dissolve the metal salt in methanol, then squirt the solution into a lit Bunsen burner from at least 1 meter away.
(B) Dissolve the metal salt in methanol, then pour the methanol into a crystallizing dish, igniting it with the flame from a Bunsen burner.
(C) Soak a wooden splint in an aqueous solution of the metal salt, then burn the splint in the flame from a Bunsen burner.
(D) Soak a wooden splint in an aqueous solution of the metal salt, then heat the splint on top of a ceramic hotplate.
64. Each of the following substances dissolves exothermically in water EXCEPT
- (A) $\text{NaOH}(s)$. (B) $\text{NH}_4\text{NO}_3(s)$.
(C) $\text{CuSO}_4(s)$. (D) $\text{H}_2\text{SO}_4(l)$.
65. Which solid is most soluble in water at 25 °C?
- (A) AgNO_3 (B) CaCO_3 (C) PbO (D) ZnS
66. A student standardizes a solution of $\text{Na}_2\text{S}_2\text{O}_3$ by titrating it against a solution containing a known mass of NaIO_3 that has been dissolved in an excess of a freshly prepared solution of KI in dilute HCl . Which of the following errors will lead to a value of the molarity of the thiosulfate solution that is higher than the true value?
- (A) The student overshoots the endpoint of the titration.
(B) The NaIO_3 is contaminated with NaCl .
(C) The KI/HCl solution is allowed to stand overnight before it is used in the titration.
(D) The sample of sodium thiosulfate pentahydrate used to make the $\text{Na}_2\text{S}_2\text{O}_3$ solution had partially dehydrated on standing.
67. Mixing equal volumes of 0.1 M $\text{Ca}(\text{NO}_3)_2$ and AgF solutions results in
- (A) no precipitate and a colorless solution.
(B) no precipitate and a colored solution.
(C) a white precipitate and a colorless solution.
(D) a colored precipitate and a colorless solution.
68. Elemental silicon is oxidized by O_2 to give a compound which dissolves in molten Na_2CO_3 . When this solution is treated with aqueous hydrochloric acid, a precipitate forms. What is the precipitate?
- (A) SiH_4 (B) SiCO_3 (C) SiO_2 (D) SiCl_4
69. A metal dissolves in 3.0 M NaOH solution with evolution of gas to form a clear, colorless solution. Upon neutralization, the solution forms a gelatinous precipitate. What is the metal?
- (A) Al (B) Ag (C) Cu (D) Mg
70. Which noble gas is most abundant in the earth's atmosphere?
- (A) He (B) Ne (C) Ar (D) Rn
71. Which of the acids below has the most ionizable hydrogen atoms per molecule?
- I. H_3PO_4 II. H_3PO_3 III. H_3PO_2
- (A) I only
(B) II only
(C) III only
(D) Each one contains the same number of ionizable H atoms.
72. Aluminum is most often found in nature as the
- (A) carbonate. (B) chloride.
(C) oxide. (D) sulfide.
73. According to the Tyndall effect, a beam of light becomes visible when passed through all of the following except a(n)
- (A) aerosol. (B) colloid.
(C) emulsion. (D) solution.
74. All of the following can be used as primary standards in acid-base titrations EXCEPT
- (A) oxalic acid.
(B) potassium hydrogen phthalate.
(C) sodium carbonate.
(D) sodium hydroxide.
75. Which of the graphs shown below would best represent the changes when 0.10 M barium hydroxide was titrated with 0.10 M sulfuric acid?
- (A)  (B) 
- (C)  (D) 

Descriptive/Laboratory

76. The arrangement of ions in a solid is best investigated by means of
- (A) infrared spectroscopy.
 (B) mass spectroscopy.
 (C) UV-visible spectroscopy.
 (D) x-ray crystallography.

77. Which substance is used in self-contained breathing equipment because it absorbs exhaled CO_2 and H_2O and releases O_2 gas?

(A) KO_2 (B) Na_2O_2 (C) NaOH (D) Li_2O

78. A sample of a white solid is known to be NaHCO_3 , AgNO_3 , Na_2S , or CaBr_2 . Which 0.1 M aqueous solution can be used to confirm the identity of the solid?

(A) $\text{NH}_3(\text{aq})$ (B) $\text{HCl}(\text{aq})$
 (C) $\text{NaOH}(\text{aq})$ (D) $\text{KCl}(\text{aq})$

79. When the permanganate ion, MnO_4^- , acts as an oxidizing agent it forms different products depending on the pH of the solution. Which species correspond to the conditions listed?

	acidic	basic	neutral
A	Mn^{2+}	$\text{Mn}(\text{OH})_2$	MnO_2
B	Mn^{2+}	MnO_4^{2-}	MnO_2
C	MnO_2	MnO_4^{2-}	$\text{Mn}(\text{OH})_2$
D	Mn^{2+}	$\text{Mn}(\text{OH})_2$	MnO_4^{2-}

(A) A (B) B (C) C (D) D

80. In an experiment to verify the value of absolute zero, a student is instructed to measure the volume of He in a 10-mL syringe at 10°C intervals between 0° and 100°C . She is told to plot the volume versus temperature and to extrapolate this graph to zero volume and read the resulting temperature. Which modification of the experimental procedure will give the best value for absolute zero?

- (A) Correcting each measured volume to one atmosphere pressure before plotting.
 (B) Doubling the number of temperature-volume values between 0° and 100°C .
 (C) Using a thermometer that can measure temperature to $\pm 0.10^\circ\text{C}$ between 0° and 100°C .
 (D) Measuring the volume of He in the syringe at -40°C and -80°C .

81. Which solid is much more soluble in 1 M HCl than in H_2O ?

(A) CaHPO_4 (B) CaCl_2
 (C) BaBr_2 (D) BaSO_4

82. Which experimental procedure is best suited to determine the H_2O_2 concentration in an aqueous solution?

(A) precipitation with standard MgCl_2 solution
 (B) reaction with excess Zn to form H_2
 (C) titration with standard H_2SO_4
 (D) titration with standard KMnO_4

83. When equal volumes of 0.2 M solutions of the following compounds are mixed, which combination forms a red precipitate?

(A) $\text{AgNO}_3 + \text{Na}_2\text{S}$ (B) $\text{AgNO}_3 + \text{K}_2\text{CrO}_4$
 (C) $\text{NiCl}_2 + \text{NaOH}$ (D) $\text{CuSO}_4 + \text{NH}_3$

84. In an experiment to determine the empirical formula of magnesium oxide, a student weighs an empty crucible then adds a strip of magnesium metal and reweighs the crucible. The crucible and magnesium are heated with a burner flame, which ignites the magnesium and forms a gray-white solid. After cooling, the crucible and solid are reweighed and the data are analyzed to give an empirical formula of Mg_5O_4 . Which could account for the observed Mg_5O_4 result rather than the expected MgO ?

- (A) Some of the magnesium reacts with atmospheric nitrogen to produce magnesium nitride.
 (B) A mixture of magnesium oxide and magnesium peroxide forms during combustion.
 (C) The piece of magnesium ribbon is shorter than recommended in the procedure.
 (D) The crucible and magnesium are heated longer than recommended in the procedure.

85. An acidic solution of methyl red has an absorbance of 0.451 at 530 nm in a 5.00 mm cell. Calculate the molarity of methyl red in this solution.

[molar absorptivity = $1.06 \times 10^5 \text{ L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$ at 530 nm]

(A) $2.13 \times 10^{-6} \text{ M}$ (B) $4.26 \times 10^{-6} \text{ M}$
 (C) $8.51 \times 10^{-6} \text{ M}$ (D) $1.05 \times 10^{-5} \text{ M}$

86. Which solid does not react with a small amount of 3 M HNO_3 ?

(A) calcium carbonate (B) manganese(II) sulfide
 (C) potassium sulfite (D) silver chloride

87. A metal dissolves in dilute HCl in the absence of air to form a pale green solution that turns yellow upon exposure to air. The metal could be

(A) iron. (B) manganese.
 (C) nickel. (D) vanadium.

Descriptive/Laboratory

88. A Material Safety Data Sheet (MSDS) provides what type(s) of information about a chemical?
I First aid measures II Handling and storage tips
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II
89. When a 25.00 mL volumetric flask weighing 20.340 g is filled partially with metal shot the mass is 119.691 g. The flask is then filled to the 25.00 mL mark with methanol ($d = 0.791 \text{ g}\cdot\text{cm}^{-3}$) and has a total mass of 130.410 g. What is the density of the metal?
(A) $6.96 \text{ g}\cdot\text{cm}^{-3}$ (B) $8.68 \text{ g}\cdot\text{cm}^{-3}$
(C) $9.27 \text{ g}\cdot\text{cm}^{-3}$ (D) $11.7 \text{ g}\cdot\text{cm}^{-3}$
90. An aqueous solution contains the ions Ag^+ , Ba^{2+} , and Ni^{2+} . Dilute aqueous solutions of NaCl , Na_2S , and Na_2SO_4 are available. In what order should these solutions be added if the goal is to precipitate each of the three cations separately?
(A) Na_2S , Na_2SO_4 , NaCl (B) Na_2S , NaCl , Na_2SO_4
(C) Na_2SO_4 , Na_2S , NaCl (D) NaCl , Na_2SO_4 , Na_2S
91. A student prepares a 100 mL aqueous solution containing a small amount of $(\text{NH}_4)_2\text{SO}_4$ and a second 100 mL solution containing a small amount of NaI , then mixes the two solutions. Which statement describes what happens?
(A) Both compounds dissolve and remain in solution when the two solutions are mixed.
(B) Both compounds dissolve initially but NH_4I precipitates when the solutions are mixed.
(C) Both compounds dissolve initially but Na_2SO_4 precipitates when the solutions are mixed.
(D) The NaI dissolves but the $(\text{NH}_4)_2\text{SO}_4$ does not. There is no change upon mixing.
92. A colored gas is observed with which combination?
(A) calcium hydride and water
(B) lead metal and nitric acid
(C) sodium carbonate and sulfuric acid
(D) zinc sulfide and hydrochloric acid
93. Mixing which pair of 0.10 M solutions produces two precipitates that cannot be separated from one another by filtration?
(A) aluminum chloride and copper(II) nitrate
(B) strontium bromide and lead(II) acetate
(C) magnesium perchlorate and lithium carbonate
(D) barium hydroxide and copper(II) sulfate
94. Which gas turns limewater, a saturated solution of $\text{Ca}(\text{OH})_2$, cloudy?
(A) H_2 (B) O_2 (C) CO_2 (D) CH_4
95. For aqueous solutions of which of the following substances could the concentration be determined by visible spectrophotometry?
I $\text{Cr}(\text{NO}_3)_3$ II KMnO_4 III $\text{Zn}(\text{NO}_3)_2$
(A) I only (B) III only
(C) I and II only (D) I, II, III
96. Four elements were tested in the laboratory and gave the results in the table below. Which element is a metalloid?
- | Element | Appearance | Conductivity | Behavior with HCl |
|---------|---------------|--------------|-------------------|
| A | Slight luster | High | Bubbles slowly |
| B | Shiny | Low | No reaction |
| C | Dull | None | No reaction |
| D | Shiny | High | Bubbles rapidly |
- (A) Element A (B) Element B
(C) Element C (D) Element D
97. Which solute has the greatest solubility (in mol/L) in water at 25°C and 1 atm?
(A) CH_4 (B) NH_3 (C) AgCl (D) CaSO_4
98. Which 2.00 M solution can be used to separate Al^{3+} from Fe^{3+} in an aqueous solution?
(A) HCl (B) H_2SO_4 (C) NaCl (D) NaOH
99. Which oxide produces an acidic solution when mixed with water?
(A) Al_2O_3 (B) CaO (C) CO (D) SO_2
100. Which gas should not be collected over water due to its high solubility in water?
(A) H_2 (B) N_2 (C) NH_3 (D) CH_4

Descriptive/Laboratory

101. Which procedure(s) will allow a student to differentiate between solid sodium sulfate and solid sodium sulfite?

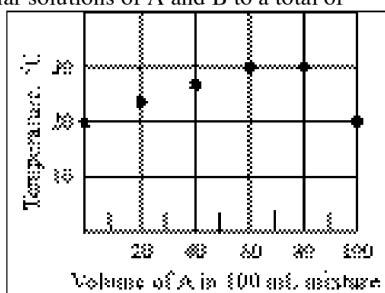
- I.** Make solutions of each and look for a precipitate when added to 0.10 M $\text{Ba}(\text{NO}_3)_2$.
II. Add crystals of each to 0.10 M HCl and watch for bubbles.
III. Make solutions of each and test with a pH indicator.

- (A) I only (B) III only
 (C) I and II only (D) II and III only

102. A student gets fingerprints on a cuvette before using it to determine the concentration of a colored species using its known extinction coefficient. What is the effect on the absorbance and reported concentration?

- | | absorbance | reported concentration |
|-----|------------|------------------------|
| (A) | increased | too low |
| (B) | increased | too high |
| (C) | decreased | too low |
| (D) | decreased | too high |

103. Chemicals A and B each initially at 20 °C react exothermically. A graph of the final temperature reached by mixing equimolar solutions of A and B to a total of 100 mL is given to the right. According to this graph, in what mole ratio do A and B react?



- (A) 2:1 (B) 3:1 (C) 4:1 (D) 3:2

104. What color does the strontium ion produce in a flame test?

- (A) red (B) blue (C) green (D) yellow

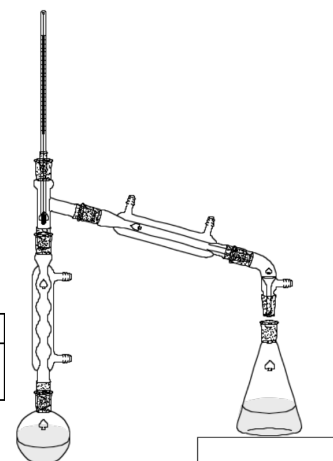
105. Which reagents produce a gas when combined?

- I.** HCl and Na_2SO_3
II. NaOH and Al

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

106. A 1:1 mixture of pentane and hexane is separated by fractional distillation in the apparatus shown. At what temperature does the first drop of condensate appear on the thermometer?

Boiling point / °C	
pentane	36
hexane	69



- (A) less than 36 °C (B) 36 °C
 (C) between 36 °C and 69 °C (D) more than 69 °C

107. Which nitrogen halide is least stable thermodynamically?

- (A) NF_3 (B) NCl_3 (C) NBr_3 (D) NI_3

108. Cyclohexane and water can be separated by using a separatory funnel. Which property contributes to this separation?

- (A) Cyclohexane and water are immiscible.
 (B) Cyclohexane has a lower viscosity than water.
 (C) Cyclohexane has a greater molar mass than water.
 (D) Cyclohexane has a greater vapor pressure than water.

109. A NaOH solution is to be standardized by titrating it against a known mass of potassium hydrogen phthalate. Which procedure will give a molarity of NaOH that is too low?

- (A) Deliberately weighing one half the recommended amount of potassium hydrogen phthalate.
 (B) Dissolving the potassium hydrogen phthalate in more water than is recommended.
 (C) Neglecting to fill the tip of the buret with NaOH solution before titrating.
 (D) Losing some of the potassium hydrogen phthalate solution from the flask before titrating.

110. Which absorbs gaseous carbon dioxide most effectively?

- (A) solid KOH (B) solid SiO_2
 (C) aqueous HCl (D) aqueous NaF

Descriptive/Laboratory

111. Which laboratory results will tell whether an unknown white solid is NaOH or NH_4NO_3 ?
- (A) NaOH is soluble in H_2O but NH_4NO_3 is not.
 (B) Aqueous NaOH turns litmus blue but NH_4NO_3 does not.
 (C) Aqueous NaOH reacts with copper metal but NH_4NO_3 does not.
 (D) NaOH gives a green flame test but NH_4NO_3 is colorless in a flame.

112. Which sets of chemicals, when mixed, produce the observation(s) listed?

Combination	Observation
I. $\text{NH}_4\text{Cl}(\text{s})$ and $\text{H}_2\text{O}(\text{l})$	endothermic
II. 9 M $\text{H}_2\text{SO}_4(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$	exothermic
III. 1M $\text{NaOH}(\text{aq})$ and 1 M $\text{HCl}(\text{aq})$	exothermic

- (A) III only (B) I and II only
 (C) II and III only (D) I, II and III
113. What happens when 6 M nitric acid is added to an aqueous solution that contains 0.1 M Cl^- and 0.1 M $\text{Ag}(\text{NH}_3)_2^+$?
- (A) A deposit of silver metal forms.
 (B) A precipitate of AgCl forms.
 (C) Chlorine gas is released.
 (D) Gaseous ammonia is released.
114. A mixture of which 0.2 M aqueous solutions will form a precipitate that dissolves in 6 M nitric acid?
- (A) $\text{Co}(\text{NO}_3)_2$ and NH_4Cl (B) $\text{Pb}(\text{NO}_3)_2$ and NaBr
 (C) $\text{Ba}(\text{NO}_3)_2$ and Na_2CO_3 (D) $\text{Al}(\text{NO}_3)_3$ and K_2SO_4
115. When a liquid is delivered from a volumetric pipet a small amount is typically retained in the tip. How should a student proceed in order to deliver the volume of liquid stated on the pipet?
- (A) Leave the small amount in the tip.
 (B) Use a pipet bulb to expel the remaining droplet.
 (C) Shake the pipet to dispense the amount left in the tip.
 (D) Draw the liquid above the line initially to compensate for the amount that remains in the tip.
116. Which method for characterizing organic compounds relies on the vibration of atoms in the compound?
- (A) infrared spectroscopy
 (B) nuclear magnetic resonance spectroscopy
 (C) UV-visible spectroscopy
 (D) X-ray diffraction

117. Which substance is NOT paired correctly with its name?
- (A) baking soda - potassium hydrogen tartrate
 (B) chalk - calcium carbonate
 (C) Epsom salt - magnesium sulfate heptahydrate
 (D) Plaster of Paris - calcium sulfate hemihydrate

118. Which acid should be stored in plastic containers rather than in glass ones?
- (A) hydrofluoric acid (B) nitric acid
 (C) phosphoric acid (D) sulfuric acid

119. Which element does NOT occur as distinct allotropes at temperatures between 0°C and 150°C ?
- (A) phosphorus (B) silicon
 (C) sulfur (D) tin

120. Which gas is odorless?
- (A) CH_4 (B) HCl (C) NH_3 (D) O_3

121. Which technique is preferred for delivering a solid into a pre-weighed beaker for weighing?
- (A) Transfer more of the reagent than is needed to the beaker. Return the excess to the bottle with a spatula.
 (B) Transfer the desired amount of solid from the reagent bottle by holding the neck of the open bottle over the beaker and tapping the bottle. Then weigh the beaker and solid.
 (C) Weigh a spatula, scoop the desired amount of solid from the bottle, transfer it to the beaker and reweigh the spatula.
 (D) Weigh a piece of filter paper, tap the neck of the bottle to transfer solid to the filter paper, weigh the filter paper and transfer the solid to the beaker.

122. Bronze is an alloy of
- (A) copper and tin (B) copper and zinc
 (C) nickel and tin (D) nickel and zinc

123. An unknown anion in solution is to be identified by adding Ag^+ and Ba^{2+} ions to separate portions of it. Which anion would produce the results listed for it? (+ indicates the presence of a precipitate)

	Ag^+	Ba^{2+}
(A) carbonate	+	-
(B) hydroxide	-	+
(C) iodide	+	-
(D) sulfide	-	-

Descriptive/Laboratory

124. Which solution produces a black precipitate when added to an aqueous copper(II) solution?

- (A) NH_3 (B) $(\text{NH}_4)_2\text{S}$
(C) K_2SO_4 (D) NaOH

125. Solutions of which ion produce a red color when vaporized in a Bunsen burner flame?

- (A) calcium (B) potassium
(C) sodium (D) zinc

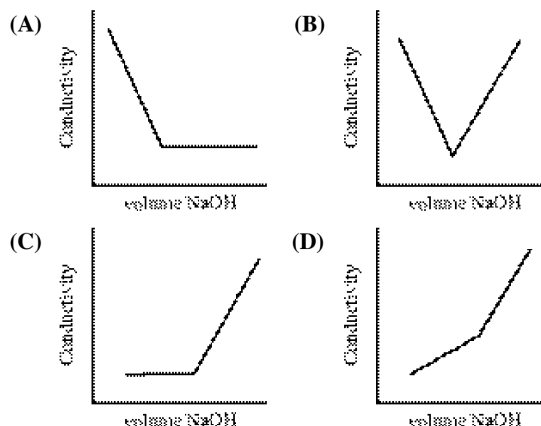
126. Which procedure for dispensing a liquid with a volumetric pipet is correct?

- (A) Draw the liquid up to the line on the pipet using a pipet bulb. Squeeze the bulb to force all the liquid in the pipet into the receiving container.
(B) Introduce the liquid into the top end of the pipet until it is filled to the line. Allow the liquid to drain into the desired container. Blow on the pipet to release the last drop.
(C) Draw the liquid above the line on the pipet using a pipet bulb. With a finger on the top of the pipet allow the curve of the meniscus to drop to the line. Place the tip of the pipet against the side of the receiving container and allow the liquid to drain.
(D) Draw the liquid above the line on the pipet by sucking on the open end of the pipet. Place a thumb on the top of the pipet and allow the curve of the meniscus to drop to the line. Allow the liquid to drain into the receiving container pipet against its side.

127. Which physical characteristic distinguishes copper from brass (an alloy of copper and zinc)?

- (A) Brass is a liquid at room temperature and copper is not.
(B) Brass is much less dense than copper.
(C) Brass is attracted to a magnet but copper is not.
(D) Brass is a much poorer electrical conductor than copper.

128. Which diagram best represents the change in electrical conductivity of a solution of acetic acid as a solution of sodium hydroxide is added?



129. Which element is obtained commercially from seawater?

- (A) bromine (B) gold
(C) iron (D) oxygen

130. Which solution can serve as both reactant and indicator when it is used in redox titrations?

- (A) $\text{FeNH}_4(\text{SO}_4)_2$ (B) KMnO_4
(C) $\text{H}_2\text{C}_2\text{O}_4$ (D) $\text{Na}_2\text{S}_2\text{O}_3$

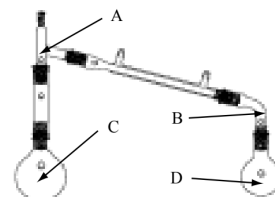
131. What is formed when a solution of NH_4NO_2 is heated gently?

- (A) N_2 and H_2O (B) N_2O and H_2O
(C) NO and H_2 (D) N_2 , H_2 and O_2

132. Which method should be used to extinguish burning magnesium metal?

- (A) Blanket it with CO_2 (B) Blow on it.
(C) Dump sand on it. (D) Pour water on it.

133. Which letter indicates where a thermometer should be placed to determine the boiling point of a distillate?



- (A) A (B) B (C) C (D) D

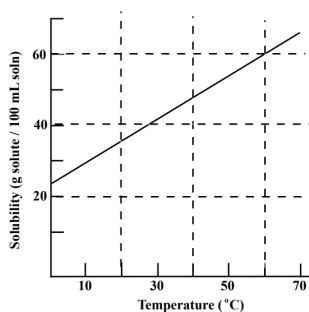
Descriptive/Laboratory

134. A 50 mL sample of gas is collected over water. What will be the effect on the calculated molar mass of the gas if the effect of the water vapor is ignored? It will be
- (A) high because of the mass of water in the collection flask.
- (B) high because of omitting the vapor pressure of the water in the calculation.
- (C) low because of the mass of water in the collection flask.
- (D) low because of omitting the vapor pressure of the water in the calculation.

135. In an experiment to determine the percentage of water in a solid hydrate by heating, what is the best indication that all the water has been removed?
- (A) The solid melts.
- (B) The solid changes color.
- (C) Water vapor no longer appears.
- (D) Successive weighings give the same mass.

136. What is the proper way to dispose of a two milliliter sample of hexane after completing experiments with it?
- (A) Return it to the solvent bottle.
- (B) Place it in a waste bottle with compatible organic materials.
- (C) Flush it down the drain with large quantities of water.
- (D) Pour it on a solid absorbent so it can be thrown away with solid waste.

137. According to the solubility curve shown, how many grams of solute can be recrystallized when 20 mL of a saturated solution at 60 °C are cooled to 0 °C?



- (A) 7.0 (B) 12 (C) 25 (D) 35

138. Which procedure is best to extinguish burning magnesium?
- (A) Add water to it.
- (B) Blow nitrogen gas over it.
- (C) Cover it with sand.
- (D) Throw ice on it.

139. Which two sets of reactants best represent the amphotericism of $\text{Zn}(\text{OH})_2$?

Set 1. $\text{Zn}(\text{OH})_2(s)$ and $\text{OH}^-(aq)$

Set 2. $\text{Zn}(\text{OH})_2(s)$ and $\text{H}_2\text{O}(l)$

Set 3. $\text{Zn}(\text{OH})_2(s)$ and $\text{H}^+(aq)$

Set 4. $\text{Zn}(\text{OH})_2(s)$ and $\text{NH}_3(aq)$

- (A) Sets 1 and 2 (B) Sets 1 and 3
- (C) Sets 2 and 4 (D) Sets 3 and 4

140. Which of these statements about sulfur is **not** correct?

- (A) It exists in different allotropic forms.
- (B) It can behave as either an oxidizing agent or a reducing agent.
- (C) It can form up to six covalent bonds in compounds.
- (D) It is a liquid at 25 °C and 1 atm pressure.

141. A weighed quantity of a gas is collected over water at 25 °C and 742 mmHg. The molar mass of the gas is to be determined at standard temperature and pressure. If the vapor pressure of water is ignored during the calculation, what is the effect on the calculated pressure and calculated molar mass of the gas?

	pressure	molar mass
(A)	low	low
(B)	low	high
(C)	high	low
(D)	high	high

142. Which of these compounds is amphoteric?

I. $\text{Al}(\text{OH})_3$ II. $\text{Ba}(\text{OH})_2$ III. $\text{Zn}(\text{OH})_2$

- (A) I only (B) II only
- (C) I and III only (D) II and III only

143. Calcium hydride reacts with excess water to form

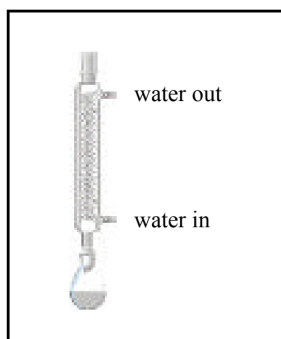
- (A) CaO and H_2 (B) $\text{Ca}(\text{OH})_2$ and O_2
- (C) $\text{Ca}(\text{OH})_2$ only (D) $\text{Ca}(\text{OH})_2$ and H_2

144. Which substance participates readily in both acid-base and oxidation-reduction reactions?

- (A) Na_2CO_3 (B) KOH
- (C) KMnO_4 (D) $\text{H}_2\text{C}_2\text{O}_4$

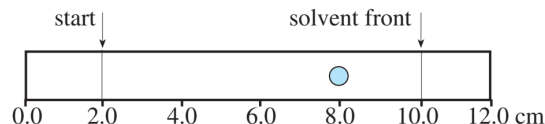
Descriptive/Laboratory

145. What is the purpose of this apparatus?



- (A) distilling (B) filtering
(C) refluxing (D) titrating
146. Which of these ions is expected to be colored in aqueous solution?
I Fe^{3+} II Ni^{2+} III Al^{3+}
(A) I only (B) III only
(C) I and II only (D) I, II, and III
147. Which substance is stored in contact with water to prevent it from reacting with air?
(A) bromine (B) lithium
(C) mercury (D) phosphorus
148. A solution of concentrated aqueous ammonia is added dropwise to 1 mL of a dilute aqueous solution of copper(II) nitrate until a total of 1 mL of the ammonia solution has been added. What observations can be made during this process?
(A) The colorless copper(II) nitrate solution turns blue and yields a dark blue precipitate.
(B) The colorless copper(II) nitrate solution yields a white precipitate which turns dark blue upon standing.
(C) The light blue copper(II) nitrate solution yields a precipitate which redissolves to form a dark blue solution.
(D) The light blue copper(II) nitrate solution turns dark blue and yields a dark blue precipitate.
149. What gas is produced when dilute HNO_3 is added to silver metal?
(A) NO (B) H_2 (C) NH_3 (D) N_2

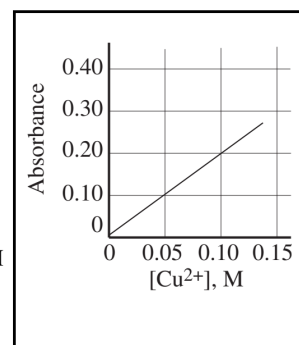
150. A substance is analyzed by paper chromatography, giving the chromatogram shown.



What is the R_f value of the substance represented by the spot at 8.0 cm?

- (A) 0.80 (B) 0.75 (C) 0.67 (D) 0.60

151. The molarity of a Cu^{2+} solution is to be determined from its absorbance, measured under the same conditions as those used to prepare this calibration curve. What will be the percent uncertainty in the concentration of a 0.050 M solution if the uncertainty in the absorbance reading is ± 0.01 absorbance units?



- (A) 5% (B) 10% (C) 15% (D) 20%

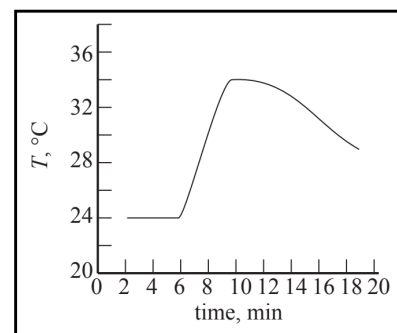
152. Which substance is most likely to be soluble in a nonpolar solvent?

- (A) glucose (B) graphite
(C) lithium fluoride (D) sulfur

153. A solution of which substance can best be used as both a titrant and its own indicator in an oxidation-reduction titration?

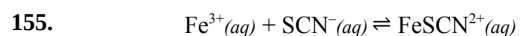
- (A) I_2 (B) NaOCl
(C) $\text{K}_2\text{Cr}_2\text{O}_7$ (D) KMnO_4

154. What value of ΔT should be used for the calorimetry experiment that gives these graphed results?



- (A) 10 °C (B) 12 °C (C) 15 °C (D) 19 °C

Descriptive/Laboratory



The equilibrium constant for this reaction can best be determined by means of

- (A) chromatography. (B) conductance.
(C) ion exchange. (D) spectrophotometry.

156. Which solid reacts with dilute hydrochloric acid at 25 °C to produce a gas that is more dense than air?

- (A) Zn (B) $\text{Pb}(\text{NO}_3)_2$
(C) NaBr (D) NaHCO_3

157. When ionic hydrides react with water, the products are

- (A) acidic solutions and hydrogen gas.
(B) acidic solutions and oxygen gas.
(C) basic solutions and hydrogen gas.
(D) basic solutions and oxygen gas.